Summary in Graph

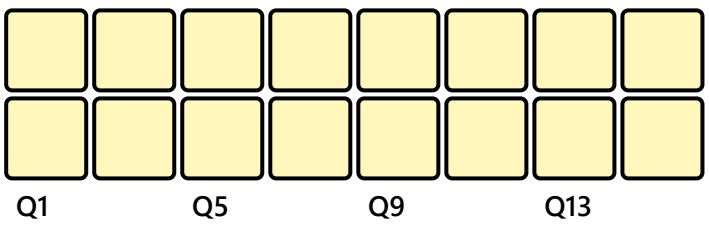
Exam Summary (GO Classes GATE CS | Propositional Logic| Weekly Quiz 2)

Qs. Attempted:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)
Correct Marks:	0 (0 + 0)
Penalty Marks:	0 (0 + 0)
Resultant Marks:	0 (0 + 0)

Total Questions:	16 (7 + 9)
Total Marks:	25 (7 + 18)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (92) out of 919 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Mathematical Logic

What is logically equivalent to "If Kareena and Parineeti go to the shopping mall then it is raining":

A. If Kareena and Parineeti do not go to the shopping mall then it is not raining.
B. If Kareena and Parineeti do not go to the shopping mall then it is raining.
C. If it is raining then Kareena and Parineeti go to the shopping mall.
D. If it is not raining then Kareena and Parineeti do not go to the shopping mall.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Which of the following expressions is false?

A. $p \rightarrow q \equiv q \rightarrow p$
B. $\neg(p \vee q) \equiv \neg p \wedge \neg q$
C. $p \rightarrow q \equiv \neg q \rightarrow \neg p$
D. none of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Select the proposition that is a tautology.

A. $(p \wedge q) \rightarrow \neg p$
B. $(p \vee q) \rightarrow p$
C. $(p \wedge q) \leftrightarrow p$
D. $(p \wedge q) \rightarrow p$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Select the proposition that is a contradiction.

A. $\neg(p \vee q) \wedge p$
B. $(p \vee q) \wedge p$
C. $(\neg p \wedge q) \leftrightarrow p$
D. $(\neg p \wedge q) \rightarrow p$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

In propositional logic, given P and $P \rightarrow Q$, we can infer _____

A. $\sim Q$
B. Q
C. $P \wedge Q$
D. $\sim P \wedge Q$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Which of the following statements is true?

A. The sentence S is a logical consequence of $S_1, \ldots, S_n$ if and only if $S_1 \wedge S_2 \wedge \cdots \wedge S_n \rightarrow S$ is satisfiable.
B. The sentence S is a logical consequence of $S_1, \ldots, S_n$ if and only if $S_1 \wedge S_2 \wedge \cdots \wedge S_n \rightarrow S$ is valid.
C. The sentence S is a logical consequence of $S_1, \ldots, S_n$ if and only if $S_1 \wedge S_2 \wedge \cdots \wedge S_n \wedge S$ is consistent.
D. The sentence S is a logical consequence of $S_1, \ldots, S_n$ if and only if $S_1 \wedge S_2 \wedge \cdots \wedge S_n \wedge S$ is inconsistent.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

If F_1, F_2 and F_3 are propositional formulae/expressions, over same set of propositional variables, such that $F_1 \wedge F_2 \rightarrow F_3$ is a contradiction, then which of the following is/are necessarily true?

A. Both F_1 and F_2 are tautologies
B. The conjunction $F_1 \wedge F_2$ is a tautology
C. F_3 is a contradiction
D. $F_1,\ F_2$ and F_3 all are contradictions.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Let φ be a propositional formula on a set of variables A and ψ be a propositional formula on a set of variables B, such that $\varphi \Rightarrow \psi$. A $Craig\ interpolant$ of φ and ψ is a propositional formula μ on variables $A \cap B$ such that $\varphi \Rightarrow \mu$ and $\mu \Rightarrow \psi$. Given propositional formula $\varphi = q \vee (r \wedge s)$ on the set of variables $A = \{q, r, s\}$ and propositional formula $\psi = \neg q \rightarrow (s \vee t)$ on the set of variables $B = \{q, s, t\}$, which of the following is a Craig interpolant for φ and ψ?

A. φ itself
B. $q \vee s$
C. $q \vee r$
D. $\neg q \wedge s$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Recall the definitions of valid, satisfiable, and unsatisfiable in the propositional logic.

What is the relationship between the above concepts?

A. If a formula ϕ is valid then ϕ is satisfiable; and if ϕ is invalid then ϕ is not satisfiable.
B. If a formula ϕ is valid then $\neg\phi$ is not satisfiable; and if ϕ is invalid then $\neg\phi$ is satisfiable.
C. If a formula $\neg\phi$ is not satisfiable, then ϕ is valid; and if $\neg\phi$ is satisfiable then ϕ is invalid.
D. If a formula ϕ is not satisfiable, then ϕ is invalid; and if $\neg\phi$ is satisfiable then ϕ is valid.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

In propositional logic if $(P \to Q) \land (R \to S)$ and $(P \lor R)$ are two premises such that

$$\frac{\begin{array}{c}(P \to Q) \land (R \to S)\\ P \lor R\end{array}}{Y}$$

Y is the premise :

A. $P \lor R$
B. $P \lor S$
C. $Q \lor R$
D. $Q \lor S$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

A set of propositions is called a system specification.

System specification is consistent if they do not contain conflicting requirements that could be used to derive a contradiction.
When specifications are not consistent, there would be no way to develop a system that satisfies all specifications.
Consider the following System Specifications:

1. "Whenever the system software is being upgraded, users cannot access the file system. If users can access the file system, then they can save new files. If users cannot save new files, then the system software is not being upgraded."
2. "The diagnostic message is stored in the buffer or it is retransmitted. The diagnostic message is not stored in the buffer. If the diagnostic message is stored in the buffer, then it is retransmitted."

Which of the above system specifications are consistent?

A. Only 1
B. Only 2
C. Both
D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Numerical Type | Award: 2 | Penalty: 0 | Mathematical Logic

Two compound propositions are logically equivalent if they have the same truth table. For example, the following two compound propositions are logically equivalent: $\mathrm{p} \to \mathrm{q}$ and $\mathrm{q} \lor \neg \mathrm{p}$. Using exactly $\mathrm{k} = 4$ propositional variables, how many compound propositions are there that are Not logically equivalent to each other?

Your Answer: | Correct Answer: 65536 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #13 Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let p, q, r be three propositional variables.

Which of the following statements is/are false?

A. $p \rightarrow (q \vee r)) \equiv ((p \wedge \neg q) \rightarrow r)$
B. $(p \wedge q) \vee r \equiv p \wedge (q \vee r)$
C. If $(p \rightarrow q)$ is FALSE then $(q \rightarrow p)$ is TRUE.
D. If $(p \rightarrow q)$ is TRUE then $(q \rightarrow p)$ is FALSE.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Consider a proposition given as:

$x \geq 6$, if $x^2 \geq 25$ and and its proof as:

If $x \geq 6$, then $x^2 = x.x \geq 6.6 = 36 \geq 25$

Which of the following is correct with respect to the given proposition and its proof?

a. The proof shows the converse
b. The proof starts by assuming what is to be shown
c. The proof is correct and there is nothing wrong

A. a only
B. c only
C. a and b
D. b only

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

- If the bank receipt is forged, then Mr. M is liable.
- If Mr. M is liable, he will go bankrupt.
- If the bank will loan him money, he will not go bankrupt.
- The bank will loan him money.

Which of the following can be concluded from the above statements?

A. Mr. M is liable
B. The receipt is not forged
C. Mr. M will go bankrupt
D. The bank will go bankrupt

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

If $\mathrm{F1}, \mathrm{F2}$ and $\mathrm{F3}$ are propositional formulae/expressions, over some set of propositional variables, such that $\mathrm{F1} \vee F2 \rightarrow \mathrm{F3}$ is a contradiction, then which of the following is/are necessarily true:

A. Both $\mathrm{F1}, \mathrm{F2}$ are tautologies.
B. At least one of $\mathrm{F1}, \mathrm{F2}$ is a tautology.
C. $\mathrm{F3}$ is a contradiction.
D. $\mathrm{F1} \vee \mathrm{F2}$ is a tautology.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff